People's Democratic Republic of Algeria
Ministry of Higher Education and Scientific Research
The National School of Artificial Intelligence
Intelligent Systems Engineering Department

الجمهورية الجزائرية الديمقراطية الشعبية
وزارة التعليم العالي والبحث العلمي
المدرسة الوطنية العليا للذكاء الاصطناعي
قسم هندسة الأنظمة الذكية

Human Computer Interaction Labs 4th Year **2026/2027**

Lab 1 Semester: S7

Introduction to Human Computer Interaction: Understanding HCI by Observing Real Interfaces (UX/UI)

In this lab, you will explore the basics of Human-Computer Interaction (HCI) by observing how design affects the way users interact with digital systems. Through real examples, you will begin to understand how UI (User Interface) and UX (User Experience) shape the success or failure of an application.

Objectives:

- Understand what HCI means and why it is important.
- Observe and analyze how users interact with real apps and websites.
- Recognize how design elements like layout, color, and navigation impact the user experience.
- Begin to identify good and bad design practices through direct interaction.
- **Work in groups of 3 max to generate more reliable results and more accurate findings related to *UI* and *UX*.**

Lab Work

1. **Activity 1: Answer the following and Warm-Up Thoughts [10mins Disc]**
   A. What makes a website or app *"easy to use"*?
   B. Can you name an app or website that frustrated you? What was the problem?
   C. In your opinion, what's the difference between *UI and UX*?
   D. After answering all these questions can you define *HCI*? What is *HCI* in your opinion?

2. **Activity 2: Observing User Interactions [15mins + 10mins Disc]**

To observe and understand how users interact with a website or an app, and to be able to identify potential UI and UX issues. We have the following websites and apps to examine to simulate user behavior **e.g.,** Navigation issues, Confusing Layout, Number of clicks to complete a task, Fonts, Colors, Intuitive,...

A. What do you think of these e-commerce websites?
   1. Arngren 2.Amazon 3.Temu 4.Aliexpress

B. Download the following app, what do you think?

C. Navigate the following Higher Education **PROGRES WebEtu** website and app. Write down the problems you faced and their potential solutions.

3. **Activity 3: Checking How User-Friendly a Website/App Feels [25mins + 10mins Disc]**

A. **Task:** Find a wireless mouse under $20, identify the cheapest suitable option, check its rating and delivery information, add it to the cart, and determine the final price.
   1. Arngren 2.Amazon 3.Temu 4.Aliexpress

   Answer the following for each website:

   - How long did it take you?
   - How many clicks did you need?
   - Where did you feel lost?
   - What attracted your attention first?
   - Did the interface try to influence your decision?
   - Did you feel pressured to buy?
   - Could you distinguish advertising from actual product information?
   - What information was difficult to find?
   - Which website made you trust the purchase most?
   - Which interface would you redesign first and why?

B. Create a playing list on *Spotify*, or on *YouTube*, or save a random reel on *Instagram*. Retrieve and Use the saved list or reel.

C. Try to find a Radialogist in the wilaya of Algiers on Atebaa AlJazayer App.
   What to Do?
   - Compare the two websites and apps, focusing on *ease of navigation, design clarity, visual appeal, and task completion efficiency.*
   - Discuss how design choices impact the overall of your UX.

4. **Activity 4: Spot issues in your Apps and websites [10mins + 10mins Disc]**

From your work on last year's Software Engineering, Mobile and Web development, spot the issues related to the UI and UX. **List them and propose some potential solutions**.